

ROXANNE LYLE

AI & DATA ANALYTICS | MACHINE LEARNING | RESPONSIBLE AI | NLP

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PROFESSIONAL SUMMARY

AI and data analytics candidate completing a B.S. in Information Technology with a specialization in Data Analytics and Artificial Intelligence and a 4.0 GPA. Builds and audits machine-learning, NLP, explainability, responsible-AI, and forecasting workflows in Python using scikit-learn, SHAP, AIF360, TensorFlow/Keras, and OpenAI APIs. Emphasizes reproducibility, cross-validation, benchmark comparison, model interpretation, and evidence-based limitations, backed by financial-services and compliance experience handling sensitive data and structured operations.

TECHNICAL EXPERTISE

Programming & Data: Python, pandas, NumPy, data cleaning, EDA, feature preparation, CSV/JSON, SQL fundamentals

Machine Learning & Predictive Modeling: scikit-learn, classification, regression, clustering, Logistic Regression, Random Forest, Gradient Boosting, cross-validation, model comparison

Explainable & Responsible AI: SHAP, AIF360, Reweighting, Adversarial Debiasing, fairness auditing, statistical parity, disparate impact, equal opportunity

NLP & Generative AI: OpenAI API, GPT-based workflows, transformer concepts, prompt evaluation, source fidelity, response-quality analysis

Deep Learning, Forecasting & Tools: TensorFlow/Keras, CNNs, reinforcement learning, Q-learning, ARIMA, Matplotlib, Jupyter Notebook, Google Colab, Microsoft Excel

SELECTED AI & DATA PROJECTS

Explainable AI & Predictive Modeling with SHAP

PYTHON | SCIKIT-LEARN | SHAP | RANDOM FOREST | CALIFORNIA HOUSING

- Built a **100-tree Random Forest** on California Housing data; held-out **R^2 0.8051**, **RMSE 0.5053**, **MAE 0.3275**; 5-fold cross-validation **R^2 0.8045 +/- 0.0068**.
- Audited global and local model behavior with **SHAP TreeExplainer**, feature-attribution analysis, and extreme-error review while separating explanation from causal or fairness claims.

Bias Detection & Mitigation in Income Prediction

PYTHON | AIF360 | LOGISTIC REGRESSION | TENSORFLOW | UCI ADULT

- Evaluated the UCI Adult dataset with Logistic Regression: **85.38% accuracy**, **ROC-AUC 0.9099**, baseline **SPD -0.1876**, and **disparate impact 0.2947**.
- Compared **Reweighting** and **Adversarial Debiasing** across SPD, DI, EOD, AOD, accuracy, and F1, documenting fairness-performance trade-offs rather than treating any one metric as a fairness verdict.

Healthcare Patient Outcome Predictive Modeling

PYTHON | SCIKIT-LEARN | RANDOM FOREST | UCI HEART DISEASE

- Developed a Random Forest classification pipeline with **88.52% test accuracy**, **ROC-AUC 0.9334**, **92.86% sensitivity**, **F1 88.14%**, and 5-fold cross-validation **ROC-AUC 0.8888**.
- Reviewed false positives and false negatives, calibration uncertainty, fairness, and deployment limitations instead of treating strong test metrics as evidence of clinical readiness.

Generative AI & Natural Language Processing Evaluation

PYTHON | OPENAI API | NLP | PROMPT & RESPONSE EVALUATION

- Designed a controlled **GPT-based NLP evaluation** using the Gettysburg Address; assessed fluency, source fidelity, instruction compliance, limitations, secure API-key handling, and reproducibility.

Time-Series Sales Forecasting & Benchmark Evaluation

PYTHON | ARIMA | SEASONAL-NAIVE BENCHMARKING | TIME-SERIES ANALYSIS

- Compared **ARIMA** with a **seasonal-naive baseline** on monthly retail sales; the **seasonal-naive method outperformed ARIMA** on the held-out evaluation window, supporting evidence-based model selection and careful interpretation of a short time series.

EDUCATION

CAPELLA UNIVERSITY | Bachelor of Science in Information Technology

Specialization: Data Analytics and Artificial Intelligence | Expected September 2028 | GPA 4.0

Relevant Coursework: Advanced Artificial Intelligence; Data Analytics and AI in the Cloud; Introduction to Artificial Intelligence; Python Programming; Relational Databases; Statistics

ADDITIONAL TECHNICAL PROJECTS

AI Planning Agent for Room Navigation

PYTHON | PLANNING GRAPHS | BREADTH-FIRST SEARCH

- Modeled room-to-room state transitions and used **breadth-first search** to extract a **4-action plan** from Room A to Room F; the executed run expanded **6 states**, generated **11 actions**, and built **9 planning levels**.

ADDITIONAL PROJECT EXPERIENCE N-Queens constraint satisfaction; Iris classification and clustering; housing regression; CIFAR-10 CNN image classification; Q-learning grid-world agent; GAN image generation.

PROFESSIONAL EXPERIENCE

TRANSFERABLE STRENGTHS Regulated data handling | documentation accuracy | compliance-oriented review | quality control | independent remote work

Working Solutions | Outbound Engagement Specialist

Glendale, AZ | May 2024 - Aug 2024

- Managed structured outbound engagement and recordkeeping in a remote, performance-driven environment while independently following established workflows and quality requirements.
- Maintained accurate activity documentation and consistent quality standards across high-volume, process-driven work.

NexRep LLC | Customer Care Representative

Peoria, AZ | Apr 2020 - Jun 2021

- Verified member information, updated records across multiple systems, and followed HIPAA-related procedures while protecting sensitive information in a remote environment.
- Resolved service issues while maintaining accurate documentation and consistent process adherence across high-volume interactions.

Aerotek | Collections Care Representative III

Phoenix, AZ | Aug 2018 - Mar 2019

- Analyzed account and delinquency information, documented payment arrangements, and applied compliance procedures in a metrics-driven financial-services environment.
- Handled sensitive financial data while maintaining accurate account records and procedural consistency.

Carver Federal Savings Bank | Universal Banker

Brooklyn, NY | Feb 2017 - May 2018

- Reviewed customer and account information, investigated discrepancies, and supported account opening, maintenance, and transaction processing.
- Maintained documentation accuracy and regulatory awareness while working with confidential financial information.

Visa Application Centre for the Consulate General of Canada | Application Analyst

New York, NY | Jan 2014 - Jan 2015

- Reviewed applications and supporting documentation for completeness, accuracy, and procedural compliance, processing sensitive records with consistent quality control and confidentiality.

ADDITIONAL FINANCIAL SERVICES & CUSTOMER OPERATIONS EXPERIENCE

JPMorgan Chase; Virtual Office Partners | 2009 - 2013

- Built foundational experience in financial transactions, customer and account records, workflow management, confidential data handling, and structured operational procedures.